POOJA KIRAN

Security Engineer | AI & Agent Security | Application Security | Cloud IAM

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Professional Experience

Independent AI Security Researcher & Engineer Aug 2024 – Present

Self-Directed Engineering | Tempe, AZ

- Built Python security tooling for MCP tool authorization, IAM policy analysis, and model artifact inspection across three independently developed projects.
- Implemented fail-closed decisions, structured findings, audit trails, and CI-based regression testing with pytest, GitHub Actions, CodeQL, and Docker.

Business & Compliance Lead | AEROSEC Aug – Dec 2025

Technology Innovation Lab Externship | Arizona State University | Externally mentored aviation project

- Defined third-party Passenger Service System and vendor API risks for a proposed airline security dashboard with compliance and incident-reporting workflows.
- Compared four commercial security solutions and modeled a proposed $120,000 year-one budget with 12-month runway; not realized funding or revenue.

Graduate Teaching Assistant | IT Grader Jan – Oct 2025

Ira A. Fulton Schools of Engineering | Arizona State University

- Assessed approximately 85 undergraduate Web Programming submissions per semester and 52 graduate cybersecurity assignments using instructor rubrics.
- Provided technical feedback on secure development, security compliance, risk analysis, and information-systems security coursework.

Security Engineering Projects

MCP Agent Security Gateway | Python | FastAPI | MCP/JSON-RPC | SIEM

- Engineered a default-deny agent-to-tool gateway with capability checks, request inspection, anti-SSRF egress rules, and rate limits.
- Implemented 69 named prompt-injection signatures, PII/exfiltration signals, hash-chained auditing, and nine Elastic Security rule records.
- Verified October 2026 Python 3.12 CI: 782 tests passed, 81.93% statement coverage; 21 core SIEM tests separately documented.

AWS Agent Identity Guard | Python | AWS IAM | SARIF 2.1.0

- Built a static workload-identity analyzer for dangerous grants, escalation paths, trust-policy weaknesses, and missing permission boundaries.
- Implemented 25 deterministic rules for wildcard access, iam:PassRole, sts:AssumeRole, and trust policies; emitted CI-readable SARIF findings.
- Verified October 2026 CI: 244 tests passed, three credential-gated tests skipped; results are static-policy checks, not live authorization.

HF Model Provenance Scanner | Python | SafeTensors | GGUF | ONNX | Keras

- Built non-executing model artifact scanning for pickle-derived, SafeTensors, GGUF, ONNX, and Keras formats.
- Combined serialization, AST/taint, provenance, and dependency checks; flagged 33 of 33 committed adversarial test fixtures.
- Verified October 2026 CI: 260 tests passed, one skipped, 75.67% statement coverage; fixtures do not establish universal detection.

Technical Skills

Programming & engineering: Python, Bash, Rust, C++; FastAPI, pytest, Docker, GitHub Actions

Security engineering: MCP/JSON-RPC, capability authorization, API security, anti-SSRF, prompt injection, threat modeling, IAM

DevSecOps & analysis: AWS IAM, SARIF 2.1.0, CodeQL, Trivy, Bandit, pip-audit, Elastic Security, SIEM, Prometheus

Education

Arizona State University | M.S., Information Technology (Security) | May 2026 | Tempe, AZ

M. S. Ramaiah University of Applied Sciences | B.Tech., Computer Science & Engineering | Aug 2023 | Bengaluru, India